

Auction Management System

MAS Final Project Documentation

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User Requirements - Auction Management System

The **Auction Management System (AMS)** is a desktop application designed to support the internal operations of a traditional auction house. The platform functions as the primary tool for staff, enabling management of all stages of the auction lifecycle, including consignment intake, cataloging, scheduling, and live sales.

The **Manager** is responsible for the overall planning and scheduling. The Manager can schedule new auctions by specifying the date and facility, assign an auctioneer to lead each sale, arrange lots in a defined order, and cancel scheduled auctions when needed.

The **Cataloguer** is responsible for receiving and preparing items for sale. The Cataloguer can register new consignors with their personal and contact details, and record consignment agreements that define the terms, such as the seller's commission rate, reserve price, and agreement validity period. For each consigned item, the Cataloguer creates a lot, capturing its title, description, condition, estimate range, and reserve price. Each lot is assigned a restriction profile that identifies any combination of: regulated (firearms, alcohol, ivory), high-value (estimate exceeding the configured threshold), or imported (cross-border provenance). The Cataloguer must complete the appropriate verification workflow for each restriction before the lot becomes eligible for auction, such as uploading regulatory licenses or customs documents. When a lot carries more than one restriction, all corresponding verifications must be completed before it becomes auction-eligible. The system allows an individual verification to be marked as pending if documents are outstanding, in which case the lot remains in the catalog but cannot be assigned to an auction. Lots from restricted or sanctioned countries are blocked from sale. If issues arise before the auction, such as authenticity disputes or withdrawn consignments, the Cataloguer can withdraw the lot from the sale.

The **Registrar** manages bidder operations. Responsibilities include registering new bidders with their personal and contact details, assigning each a permanent system identifier for future auctions, and issuing paddle numbers before each auction. The Registrar can record absentee bids on behalf of bidders unable to attend in person, specifying the maximum amount for each lot. The Registrar assigns each bidder a membership tier (Standard, Plus, or VIP) recorded on their account. Bidders who violate auction rules may be banned from future participation.

The **Auctioneer** is a staff member who conducts the sale on the floor. The Auctioneer opens each auction round to begin bidding on the current lot, records live floor bids by entering the paddle number and bid amount, and closes the round by marking the lot as sold to the highest qualifying bidder or as passed if the reserve is not met or no bids are placed. A bidder qualifies for a lot only if they have a paddle number issued for that auction and are not banned. In exceptional cases, the Auctioneer may cancel a round, voiding all bids.

Bidders and **Consignors** are not system users but are represented as first-class records. Bidders are tracked across all auctions they participate in, and consignors are tracked across all consignment agreements and lots submitted. This persistent tracking enables the system to maintain a comprehensive operational history, including all bids and lots, without requiring external parties to log into the application.

Use Case Diagram

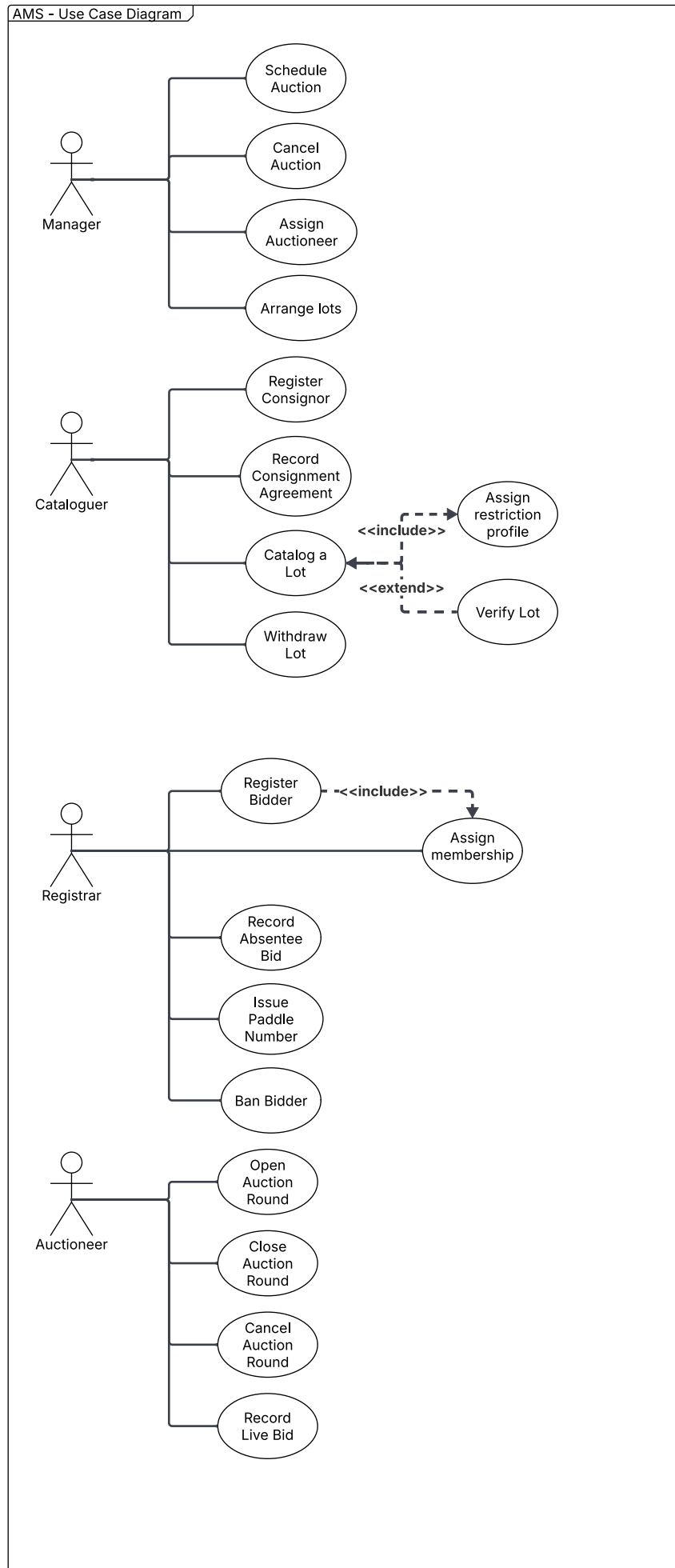


Figure 1: Auction Management System - Use Case Diagram.

Non-functional Requirements

1. A lot with any pending verification cannot be assigned to an auction.
2. A lot with a restricted/sanctioned country is blocked from sale and can never become auction-eligible.
3. Cancelling an auction round voids all bids placed in that round atomically (no partial state)
4. On equal bid amounts, a live floor bid takes precedence over an absentee bid. Between two absentee bids of equal maximum, the earlier-recorded bid wins.
5. A bidder qualifies for a lot only if they hold a paddle number for that auction, and are not banned.
6. Each bidder is assigned a permanent system identifier that persists across all auctions and is never reused.
7. All bids, lots, invoices, and payout statements are persisted. Bidders and consignors are tracked across all auctions without being system users.
8. Each staff role (Manager, Cataloguer, Registrar, Auctioneer) can access only the operations defined for that role.
9. Bidders and consignors cannot log into the application. They exist only as first-class records.
10. Configurable values - the high-value threshold, the regulated-category list, the sanctioned-country list, and tier privileges are changeable through configuration without code changes.
11. Retrieving an auction together with its lots, and each lot together with its bids, uses $O(1)$ database operations (no per-object queries).
12. Application windows are consistent with one another, and the interface helps users avoid typical input errors.
13. Live-round auctions (recording a floor bid, applying the next increment, closing a round) are operable quickly under time pressure.

Analytical Class Diagram

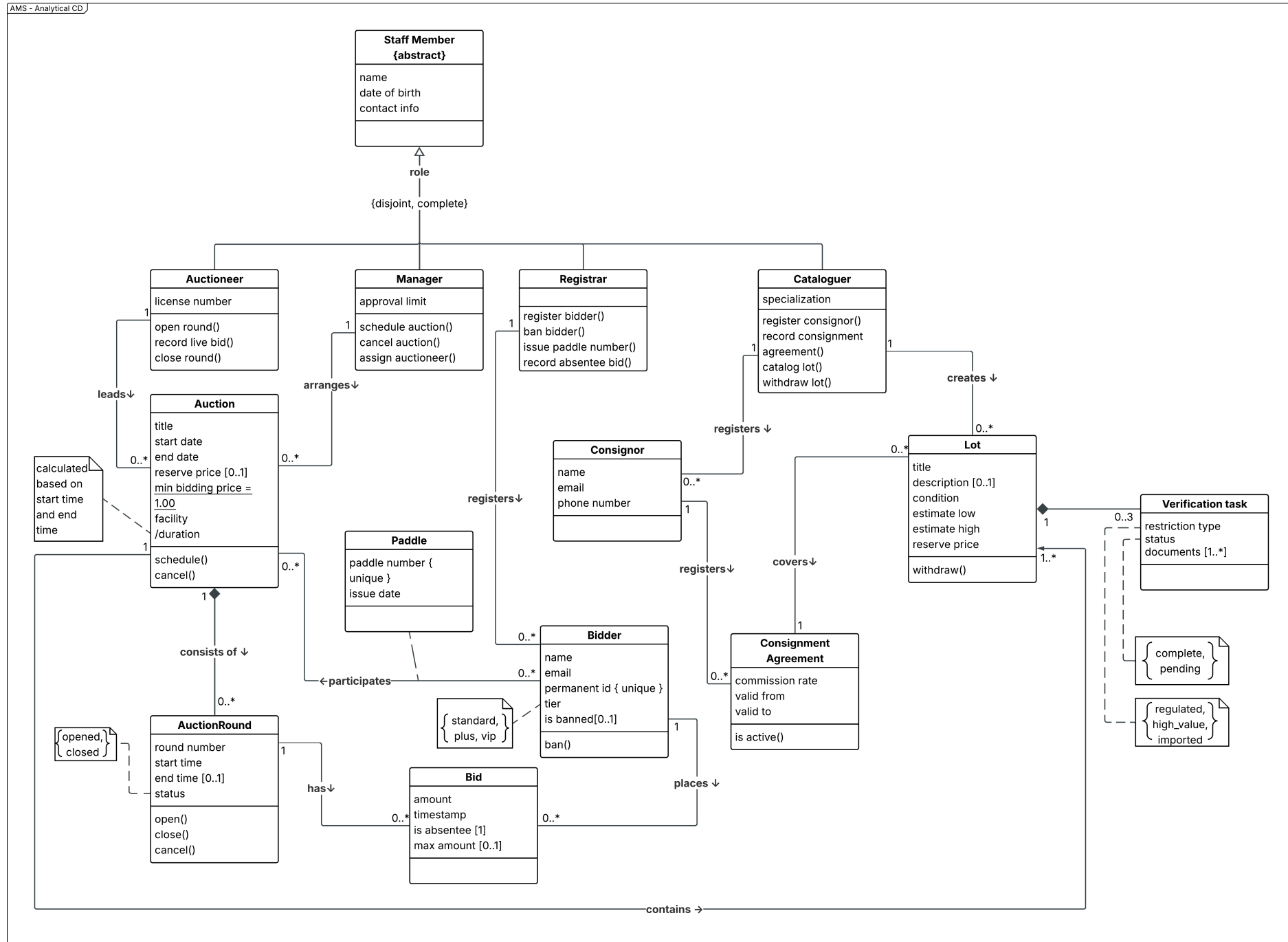


Figure 2: Auction Management System - Analytical Class Diagram

AMS - Design CD



Non-Trivial Use Case Scenario

Use Case: Catalog a Lot for Auction

Actors:

1. Cataloguer (primary)

Purpose and context:

A Cataloguer records a newly consigned item as a lot in the system, captures its details, links it to an active consignment agreement and category, completes the verification workflow for every applicable restriction, and makes it available to be scheduled into an upcoming auction - unless a restriction blocks it from sale.

Assumptions:

1. There is always at least one active consignment agreement available to attach to the lot.
2. There is always at least one category Cataloguer can assign lot to.
3. The high-value threshold, the list of restricted regulatory categories, and the restricted/sanctioned-country list are configured in the system.
4. Any documentation required for a restriction is either available to the Cataloguer or the corresponding verification is marked as “Pending”.

Pre-conditions:

1. The Cataloguer is registered and logged in.
2. The Consignor exists and holds at least one active consignment agreement.

Initiating business event: The Cataloguer clicks on the “Catalog a Lot” tab.

Basic flow of events:

1. The Cataloguer clicks on the “Catalog a Lot” tab.
2. The system displays a “Catalog a Lot” screen with a list of active consignment agreements and categories.
3. The Cataloguer selects a consignment agreement and a category, enters the lot details (e.g title, description, condition, estimate low, estimate high, reserve price), and clicks the “Submit lot” button.
4. The system validates that all required fields are populated and check price consistency (reserve price > 0, estimate high >= estimate low).
5. The system evaluates the lot’s category, estimate high, and country of origin against the configured high-value threshold, regulated-categories list, and sanctioned-countries list, and builds a restriction profile containing any combination of: **REGULATED**, **HIGH_VALUE**, **IMPORTED**, **BLOCKED**.
6. The system verifies that the restriction profile contains no active safety flags.
7. The system creates new Lot record with an “Eligible” status, persists it to the database, and displays a success confirmation message.
8. Use case ends.

Alternative flow of events:

3a. Invalid Details Entered:

3a.1. - The system detects missing required fields or inconsistent pricing (e.g. reserve price <= 0, or estimate high < estimate low).

3a.2. - The system displays a specific error message “Lot details are invalid”.

3a.3. - The system returns to step 2, preserving the Cataloguer’s entered data for correction.

3b. User Cancels Operation:

3b.1. - The Cataloguer selects to cancel operation at any point before final submission.

3b.2. - The system discards all entered inputs without persisting any records.

3b.3. - Use case ends.

6a. Lot Requires Verification:

6a.1. - The system detects that the assigned profile contains one or more of REGULATED, HIGH_VALUE, or IMPORTED (and does not contain BLOCKED).

6a.2. - The system executes Verify Lot use case to handle document uploads and tracking workflows.

6a.3. - Verify Lot executes and returns control to this point with one of three outcomes:

- “**Complete**” – continue to step 6a.4.
- “**Pending**” – proceed to alternative flow 6b.
- “**Cancelled**” – proceed to alternative flow 3b.

6a.4. - The system creates the Lot with an “Eligible” status, persists it to the database, displays a success confirmation message.

6a.5. - Use case ends.

6b. Verification Outstanding:

6b.1. - The system determines that at least one verification task was left in a “Pending” state.

6b.2. - The system creates the Lot record with a “PendingVerification” status, saving the progress but preventing it from being scheduled into auctions.

6b.3. - The system displays a confirmation message indication the lot is saved but requires outstanding verification.

6b.4. - Use case ends.

6c. Sanctioned Country Block Triggered:

6c.1. - The sub-process determines that a sanctioned country flag cannot be bypassed.

6c.2. - The system blocks standard creation process, assigns Lot state to “Blocked”, and saves the record strictly for record-keeping and audit trailing.

6c.3. - The system displays a critical security warning: “Lot from restricted/sanctioned country is blocked from sale”.

6c.4. - Use case ends.

Post condition:

Success: A new Lot exists, linked to one consignment agreement and one category, with all applicable verifications recorded. Its status is:

- **Eligible** - all verifications complete (or none required). The Lot may be assigned to an auction;
- **PendingVerification** - at least one verification pending. The Lot remains in the catalog, but is not auction-eligible;

- **Blocked** - a sanctioned-country restriction applies. The Lot is retained for record-keeping but can never become auction-eligible.

Failure / cancel: No lot is created. System state is unchanged.

Use Case “Verify Lot” Scenario

Use Case: Verify Lot

Actors:

1. Cataloguer (primary)

Purpose and context:

The Cataloguer uploads and validates the documentation required to satisfy the active restriction flags carried by a lot. The use case extends Catalog a Lot when the lot's restriction profile contains one or more of **REGULATED**, **HIGH_VALUE**, or **IMPORTED**.

Assumptions:

1. The list of acceptable document types for each restriction flag is configured in the system.
2. The list of approved third-party appraisers and the list of approved secure storage locations are configured.
3. The system provides upload functionality with basic file-level validation (file type, file size)
4. Required documentation may be partially or fully unavailable, in which case the corresponding verification task is marked as “Pending”.

Pre-conditions:

1. The primary use case “Catalog a Lot” is in progress and has reached step *6a.2*.
2. The lot's restriction profile contains at least one active restriction flag.

Initiating business event: Catalog a Lot invokes this use case at step *6a.2*. because the lot's restriction profile contains active flags.

Basic flow of events:

1. The system renders a verification panel showing one document submission section per active flag in the lot's restriction profile.
2. For each active flag, the Cataloguer uploads the required documentations:
 - **REGULATED** - uploads the regulatory permit or license and records the issuing authority and validity dates.
 - **HIGH_VALUE** - uploads a third-party appraisal document, attaches an insurance certificate covering at least the lot's estimate high, and selects a secure storage location from the approved list.
 - **IMPORTED** - uploads the provenance chain documents and enters the customs clearance import declaration number.
3. The Cataloguer clicks “Submit Verifications”
4. The system validates that all uploaded documents are technically sound.
5. The system marks each individual verification task state as “Complete”
6. Use case ends, returning the outcome “Complete” to Catalog a Lot step *6a.3*.

Alternative flow of events: **2a. Document validation fails:** **2a.1.** - The system detects that an uploaded document is malformed, or that its metadata is inconsistent.

2a.2. - The system displays an error message identifying each failed document.

2a.3. - Return to step 2 for the Cataloguer to correct or re-upload.

2b. Documents not yet available:

2b.1. - The Cataloguer indicates that one or more required documents are outstanding.

2b.2. - The Cataloguer marks the corresponding verification tasks as “Pending” and leaves additional small note on what is missing.

2b.3. - The system records those tasks with “Pending” state.

2b.4. - Use case ends, returning the outcome “Pending” to Catalog a Lot at its step 6a.3.

3a. Cataloguer cancels verification:

3a.1. - The Cataloguer chooses to cancel the verification step before submitting any documents.

3a.2. - The system discards any uploaded documents.

3a.3. - Use case ends, returning the outcome “Cancelled” to Catalog a Lot at its step 6a.3.

Post condition:

- **Complete** - all applicable verification tasks are marked “Complete”; control returns to Catalog a Lot which will set the lot’s status to “Eligible”
- **Pending** - at least one verification task is marked “Pending”; control returns to Catalog a Lot which will set the lot’s status to “PendingVerification”
- **Cancelled** - no documents are persisted; control returns to Catalog a Lot which follows its 3b. cancellation path and no lot is created.

Activity Diagram

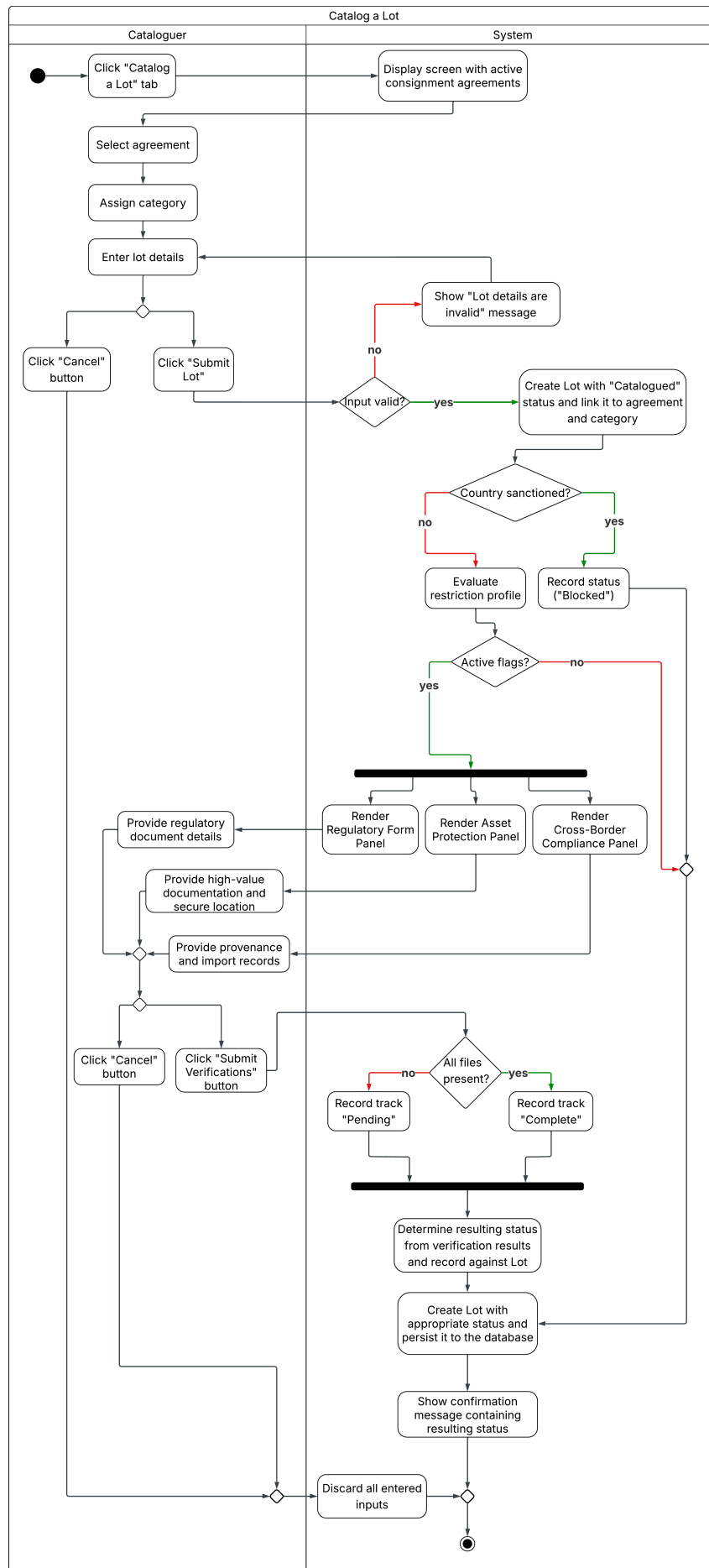


Figure 4: Auction Management System - Activity Diagram

State Diagram

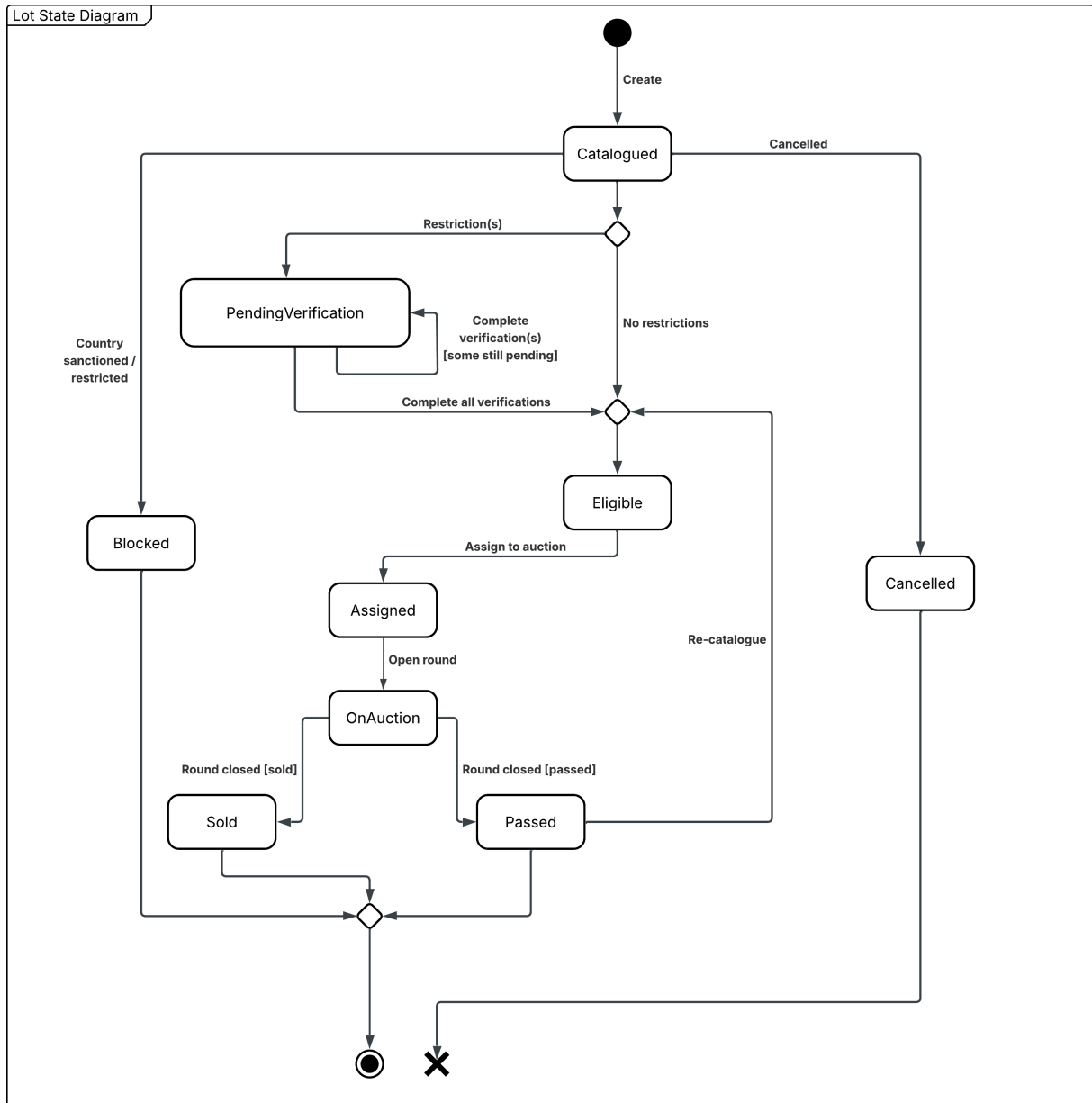


Figure 5: Auction Management System - State Diagram

GUI Design

Here is a Graphical User Interface(GUI) for a desktop Auction Management System(AMS) application.

Catalog a Lot Tab

Adam Adam Cataloguer Auction List History Catalog Catalog a Lot Analytics ⚙️📞📄	Catalog a Lot for Auction	
	Consignment Agreements	Category
● ConsignmentAgreementExample1 ● ConsignmentAgreementExample2 ● ConsignmentAgreementExample3 ● ConsignmentAgreementExample4 ● ConsignmentAgreementExample5 ● ConsignmentAgreementExample6 ● ConsignmentAgreementExample7 ● ConsignmentAgreementExample8 ● ConsignmentAgreementExample9	● CategoryExample1 ● CategoryExample2 ● CategoryExample3 ● CategoryExample4 ● CategoryExample5 ● CategoryExample6	Title Enter title: Description Enter description: Condition Enter condition: Estimate Low Enter estimate low: Estimate High Enter estimate high: Reserve Price Enter reserve price: Country of Origin Enter country of origin:
Selected: ConsignmentAgreementExample3Selected: CategoryExample2		CancelSubmit Lot

Figure 6: Auction Management System - “Catalog a Lot” Tab

Main cataloging form (empty state) The “Catalog a Lot” screen is split into 3(three) panels: a list of active consignment agreements on the left, a category selector in the middle, and lot details input form. The Cataloguer selects a consignment agreement and a category before filling in the lot fields (title, description, condition, estimate low/high, reserve price, country of origin). The bottom bar reflects the current selections.

Invalid Lot Details Error

Adam Adam Cataloguer

Auction List

History

Catalog

Catalog a Lot

Settings

Help

Export

Selected: ConsignmentAgreementExample3

Selected: CategoryExample2

Cancel

Submit Lot

Error!

Lot details are invalid

Estimate high < estimate low

Catalog a Lot for Auction

Consignment Agreements	Category	Lot Details
- ConsignmentAgreementExample1 - ConsignmentAgreementExample2 - ConsignmentAgreementExample3 - ConsignmentAgreementExample4 - ConsignmentAgreementExample5 - ConsignmentAgreementExample6 - ConsignmentAgreementExample7 - ConsignmentAgreementExample8 - ConsignmentAgreementExample9	- CategoryExample1 - CategoryExample2 - CategoryExample3 - CategoryExample4 - CategoryExample5 - CategoryExample6	Title Something Description Some description Condition Low Estimate Low 222.22 Estimate High 199.99 Reserve Price 122.3 Country of Origin Turkey

Figure 7: Auction Management System - Invalid Lot Details Error

Validation error: inconsistent prices When the Cataloguer submits a lot where estimate high is lower than estimate low, the system blocks the submission with an specific error dialog.

Restricted Country Lot Error

Adam Adam Cataloguer

Auction List

History

Catalog

Catalog a Lot

Settings

Help

Export

Selected: ConsignmentAgreementExample3

Selected: CategoryExample2

Cancel

Submit Lot

Error!

Lot was blocked from sales

Country of Origin of this Lot is on our auction's sanctioned/restricted list

Catalog a Lot for Auction

Consignment Agreements	Category	Lot Details
- ConsignmentAgreementExample1 - ConsignmentAgreementExample2 - ConsignmentAgreementExample3 - ConsignmentAgreementExample4 - ConsignmentAgreementExample5 - ConsignmentAgreementExample6 - ConsignmentAgreementExample7 - ConsignmentAgreementExample8 - ConsignmentAgreementExample9	- CategoryExample1 - CategoryExample2 - CategoryExample3 - CategoryExample4 - CategoryExample5 - CategoryExample6	Title Something Description Some description Condition Low Estimate Low 222.22 Estimate High 199.99 Reserve Price 122.3 Country of Origin Iran

Figure 8: Auction Management System - Restricted Country Lot Blocked

Blocked lot: sanctioned country When the country of origin entered is a sanctioned country, the system displays a critical error. In such a case lot is no longer auction-eligible.

Regulated Restriction Profile

Adam Adam Cataloguer

Catalog a Lot for Auction

Consignment Agreements	Category	Lot Details
● ConsignmentAgreementExample1 ● ConsignmentAgreementExample2 ● ConsignmentAgreementExample3 ● ConsignmentAgreementExample4 ● ConsignmentAgreementExample5	● CategoryExample1 ● CategoryExample2 ● CategoryRegulatedExample3 ● CategoryExample4 ● CategoryExample5	Title Something

Warning!

There is 1 restricted flag on this lot:

Regulated

Regulatory permit or license
 No file chosen

Issuing Authority

Validity Date

☐ Mark as "Pending"

High-Value

Everything is fine here ✓

Imported

Everything is fine here ✓

Cancel **Submit Verifications**

Selected: ConsignmentAgreementExample3 **Selected:** CategoryExample2

Cancel **Submit Lot**

Figure 9: Auction Management System - Regulated Documents Required

Verification dialog: one regulated flag When the selected category is regulated, a warning dialog appears showing one active restriction flag. The Regulated column displays the required document fields, while High-Value and Imported show a green checkmark confirming no action is needed there.

High-Value Restriction Profile

Adam Adam Cataloguer

Catalog a Lot for Auction

Consignment Agreements	Category	Lot Details
● ConsignmentAgreementExample1 ● ConsignmentAgreementExample2 ● ConsignmentAgreementExample3 ● ConsignmentAgreementExample4 ● ConsignmentAgreementExample5	● CategoryExample1 ● CategoryExample2 ● CategoryRegulatedExample3 ● CategoryExample4 ● CategoryExample5	Title Something

Warning!

There is 1 restricted flag on this lot:

Regulated

Everything is fine here ✓

High-Value

Third-party Appraisal Document
 No file chosen

Insurance Certificate
 No file chosen

Storage Location

☐ Mark as "Pending"

Imported

Everything is fine here ✓

Cancel **Submit Verifications**

Selected: ConsignmentAgreementExample3 **Selected:** CategoryExample2

Cancel **Submit Lot**

Figure 10: Auction Management System - High-Value Documents Required

Verification dialog: one high-value flag When the threshold triggers the high-value restriction, a warning dialog appears showing one active restriction flag. The High-Value

column displays the required document fields, while Regulated and Imported show a green checkmark confirming no action is needed there.

Imported Restriction Profile

The screenshot shows the 'Catalog a Lot for Auction' interface. A warning dialog is open, titled 'Warning!', with a close button 'X'. The dialog contains the text 'There is 1 restricted flag on this lot:'. Below this, there are three sections: 'Regulated' with the text 'Everything is fine here' and a green checkmark; 'High-Value' with the text 'Everything is fine here' and a green checkmark; and 'Imported' with the text 'Provenance Chain Documents' and a 'Choose File' button. Below this, there is a 'Customs Clearance Import Declaration Number' field with the placeholder 'Enter CIDN' and a checkbox 'Mark as "Pending"'. At the bottom right of the dialog are 'Cancel' and 'Submit Verifications' buttons. The background interface shows a table with columns 'Consignment Agreements', 'Category', and 'Lot Details'. The 'Consignment Agreements' column lists five examples, the 'Category' column lists five examples, and the 'Lot Details' column has a 'Title' field with the value 'Something'. At the bottom of the interface, there are 'Cancel' and 'Submit Lot' buttons.

Catalog a Lot for Auction		
Consignment Agreements	Category	Lot Details
- ConsignmentAgreementExample1 - ConsignmentAgreementExample2 - ConsignmentAgreementExample3 - ConsignmentAgreementExample4 - ConsignmentAgreementExample5	- CategoryExample1 - CategoryExample2 - CategoryRegulatedExample3 - CategoryExample4 - CategoryExample5	Title Something

Warning!

There is 1 restricted flag on this lot:

Regulated

Everything is fine here ✓

High-Value

Everything is fine here ✓

Imported

Provenance Chain Documents

Choose File No file chosen

Customs Clearance Import Declaration Number

Enter CIDN

☐ Mark as "Pending"

Cancel

Submit Verifications

Cancel

Submit Lot

Figure 11: Auction Management System - Imported Documents Required

Verification dialog: one imported flag When the imported restriction is triggered, a warning dialog appears showing one active restriction flag. The Imported column displays the required document fields, while Regulated and High-Value show a green checkmark confirming no action is needed there.

All Restriction Profiles

Adam Adam
Cataloguer

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Catalog a Lot for Auction

Consignment Agreements	Category	Lot Details
● ConsignmentAgreementExample1 ● ConsignmentAgreementExample2 ● ConsignmentAgreementExample3 ● ConsignmentAgreementExample4 ● ConsignmentAgreementExample5	● CategoryExample1 ● CategoryExample2 ● CategoryRegulatedExample3 ● CategoryExample4 ● CategoryExample5	Title Something

Warning!

There are 3 restricted flags on this lot:

Regulated

Regulatory permit or license

Choose file No file chosen

Issuing Authority

Some description

Validity Date

01-01-0001

☐ Mark as "Pending"

High-Value

Third-party Appraisal Document

Choose file No file chosen

Insurance Certificate

Choose file No file chosen

Storage Location

Choose location

☐ Mark as "Pending"

Imported

Provenance Chain Documents

Choose file No file chosen

Customs Clearance Import Declaration Number

Enter CIDN

☐ Mark as "Pending"

Cancel

Submit Verifications

Selected: ConsignmentAgreementExample3

Selected: CategoryRegulatedExample3

Cancel

Submit Lot

Figure 12: Auction Management System - Regulated High-Value Imported Documents Required

Verification dialog: all three flags active When all three restrictions apply simultaneously, the dialog header reads “There are 3 restricted flags on this lot” and all three columns - Regulated, High-Value and Imported - display their respective required fields at the same time. Each column has its own “Mark as Pending” checkbox, allowing the Cataloguer to handle each verification independently and it is possible not to upload all documents at once (if not ready yet).

Uploaded Documents or Data Invalid Error

The screenshot shows the 'Catalog a Lot for Auction' interface. The top bar includes the user 'Adam Adam Cataloguer' and three main sections: 'Consignment Agreements', 'Category', and 'Lot Details'. The 'Consignment Agreements' section lists five items, 'Category' lists five items, and 'Lot Details' has a 'Title' field with the value 'Something'. A 'Warning!' dialog box is open, stating 'There is 3 restricted flags on this lot:'. It contains three sections: 'Regulated' with a 'Regulatory permit or license' field (highlighted in red) containing 'RegFile.xm', 'High-Value' with a 'Third-party Appraisal Document' field (highlighted in red) containing 'TPaDD.ccs', and 'Imported' with a 'Provenance Chain Documents' field (highlighted in red) containing 'PCD982.sas'. Below these are fields for 'Issuing Authority' (YUTI-2985), 'Validity Date' (01-01-2022), and a 'Mark as "Pending"' checkbox. An 'Error!' dialog box is also open, stating 'Uploaded documents are invalid' and 'Regulatory permit document's metadata is invalid'. At the bottom right of the warning dialog are 'Cancel' and 'Submit Verifications' buttons. At the bottom right of the error dialog are 'Cancel' and 'Submit Lot' buttons.

Figure 13: Auction Management System - Invalid Documents or Data Uploaded

Verification error: invalid documents or data uploaded After the Cataloguer uploads documents and data, and clicks “Submit Verifications” button, the system validates each file. If a document fails validation, an error dialog appears highlighting the specific failure. The problematic field(s) is/are highlighted in red, allowing the Cataloguer to correct just the that document

Eligible Lot Created

The screenshot shows the 'Catalog a Lot for Auction' interface. The top bar includes the user 'Adam Adam Cataloguer' and three main sections: 'Consignment Agreements', 'Category', and 'Lot Details'. The 'Consignment Agreements' section lists nine items, 'Category' lists six items, and 'Lot Details' has fields for 'Title' (Something), 'Description' (Some description), 'Condition' (Low), 'Estimate Low' (222.22), 'Estimate High' (199.99), 'Reserve Price' (122.3), and 'Country of Origin' (Iran). A 'Success!' dialog box is open, stating 'Lot was successfully created' and 'Lot status "Eligible" meaning it can without any restrictions proceed further to be assigned to an auction'. At the bottom right of the success dialog are 'Cancel' and 'Submit Lot' buttons.

Figure 14: Auction Management System - Eligible Lot Created

Lot created notification After choosing appropriate consignment agreement, category and entering valid data - the Lot with “Eligible” status is created in the system, showing appropriate notification.

Outstanding Lot Created

The screenshot displays the 'Catalog a Lot for Auction' interface. On the left is a sidebar with the user profile 'Adam Adam Cataloguer' and navigation links: 'Auction List', 'History', 'Catalog', and 'Catalog a Lot' (which is highlighted). The main area is divided into three sections: 'Consignment Agreements' (listing 9 examples), 'Category' (listing 6 examples), and 'Lot Details' (with input fields for Title, Description, Condition, Estimate Low, Estimate High, Reserve Price, and Country of Origin). A large warning dialog box is centered on the screen, stating: 'Warning! Lot was created, but stays outstanding. Lot status "PendingVerification" meaning it is catalogued, but cannot be assigned to any auction until all necessary documents / data will be uploaded.' At the bottom, there are 'Selected' dropdowns for 'ConsignmentAgreementExample3' and 'CategoryExample2', and 'Cancel' and 'Submit Lot' buttons.

Figure 15: Auction Management System - Outstanding (PendingVerification) Lot Created

Lot created notification - PendingVerification The Cataloguer chooses appropriate consignment agreement, category and enters valid data, but Lot cannot be created because of restriction flags given to it - additional documents and data needed, but Cataloguer does not have it yet, marking lot as “Pending” - the Lot with “PendingVerification” status is created in the system.

Dynamic Analysis

Lot

Lot needed a status attribute. The status was added as an enumerated attribute with the values: *Draft*, *Eligible*, *PendingVerification*, *Blocked*, *Assigned*, *Sold*, *Passed*, *Cancelled*.

The OnAuction -> Sold and OnAuction -> Passed transitions confirmed that closing a round must update the Lot's status accordingly.

Findings from the Catalog a Lot activity diagram

The activity diagram showed even more than State Diagram - it traced the internal steps of cataloging in detail, especially the restriction and verification sub-process.

Lot missed ***countryOfOrigin*** attribute: activity diagram's step 5 shows the system evaluates the lot's country of origin against the sanctioned-restricted-country list to decide whether to assign the BLOCKED flag.

areAllVerificationsComplete() method - the activity diagram's merge point - where all three verification branches (regulated, high-value, imported) rejoin before the final status is determined.

evaluateRestrictionProfile() method - the activity diagram's evaluation step (comparing lot attributes against system configuration to build the restriction profile) implied a method for determining which restrictions apply. It checks estimate high against the threshold, the category against the regulated list, and the country against the sanctioned list.

Design Decision

Design (implementation) class diagram was created to be able to translate constructs from UML to C#/.NET.

Enumerations instead of strings

Several attributes have a fixed, closed set of valid values. There could potentially appear many typos and invalid assignments at runtime if using strings. By using C# enum types we prevent this. List of enumerations presented in the system:

- LotStatus -> Catalogued, Eligible, PendingVerification, Blocked, Assigned, OnAuction, Sold, Passed, Cancelled
- VerificationStatus -> Complete, Pending
- RestrictionType -> Regulated, HighValue, Imported
- MembershipTier -> Standard, Plus, VIP
- RoundStatus -> Opened, Closed

Abstract class and inheritance

StaffMember is declared **abstract** in C# - it cannot be instantiated directly. Four concrete child classes (subtypes): Manager, Cataloguer, Registrar, Auctioneer. Each inherit the base attributes, while some add their own specific attributes and methods. {disjoint, complete} constraint - every staff member object is exactly one of the four subtypes.

Paddle - association class

In the analytical class diagram, Paddle is an association class on the Bidder <-> Auction many-to-many relationship. C# has no concept of an association class, so that's why we transform Paddle class into a regular class with two references:

Paddle <-> Bidder - the bidder this paddle was issued to

Paddle <-> Auction - the auction this paddle is for

The Bidder-Auction many-to-many is navigated through the Paddle objects. A validation check must ensure that the same Bidder cannot receive two Paddles for the same Auction.

{unique}

Bidder permanentId attribute must be unique across all bidders. This is enforced in the property setter in C# going through all bidders searching for the desired Id. If found - exception should be thrown, otherwise Id can be assigned.

Derived attribute

Auction duration attribute is derived from StartDate and EndDate. In C# this is a read-only computer property with no backing field. No setter exists, the value is always freshly computed on read.

Bidirectional associations -> reverse-connection guards

Several associations are navigable in both directions. For example, Auction <-> AuctionRound, Bidder <-> Bid. In C# we can use .Contains() method to check whether reverse connection already exists which breaks infinite loop.

Composition -> anti-sharing enforcement

The compositions in the model (Auction <-> AuctionRound, Lot <-> VerificationTask) mean each part can belong to exactly one whole. If AuctionRound objects could freely move between Auctions, composition semantics would be violated.

In C# this is enforced with a static HashSet that tracks owned objects.

Business rules

- Lot's `assignToRound()` method - must check whether `Status == LotStatus.Eligible` before assigning
- Lot's `areAllVerificationsComplete()` method - return true only when every verification task associated with the lot has `Status == VerificationStatus.Complete`.
- Auction's `.Open()` method - checks the round's Lot is Assigned before opening bidding.

VerificationTask documents

The `documents` attribute on `VerificationTask` is multi-valued - a single verification can have several documents. In C# this is `private List<Document> _documents`, where `Document` is a custom class containing: `Id`, `FileName`, `ContentType`, `FilePath`, `UploadedType`, `Type`. Also enumeration was created for `Type` attribute `DocumentType`: `RegulatoryPermit`, `RegulatoryLicense`, `ThirdPartyAppraisal`, `InsuranceCertificate`, `ProvenanceChainDocument`.